Interior-Point Methods Strike Back: Solving the Wasserstein Barycenter Problem

the Matrix-based Adaptive Alternating IPM(**MAAIPM**)

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Outline

1. Introduction
2. MAAIPM Framework
3. Matrix Based Acceleration
4. Numerical Results

Wasserstein Barycenter Application

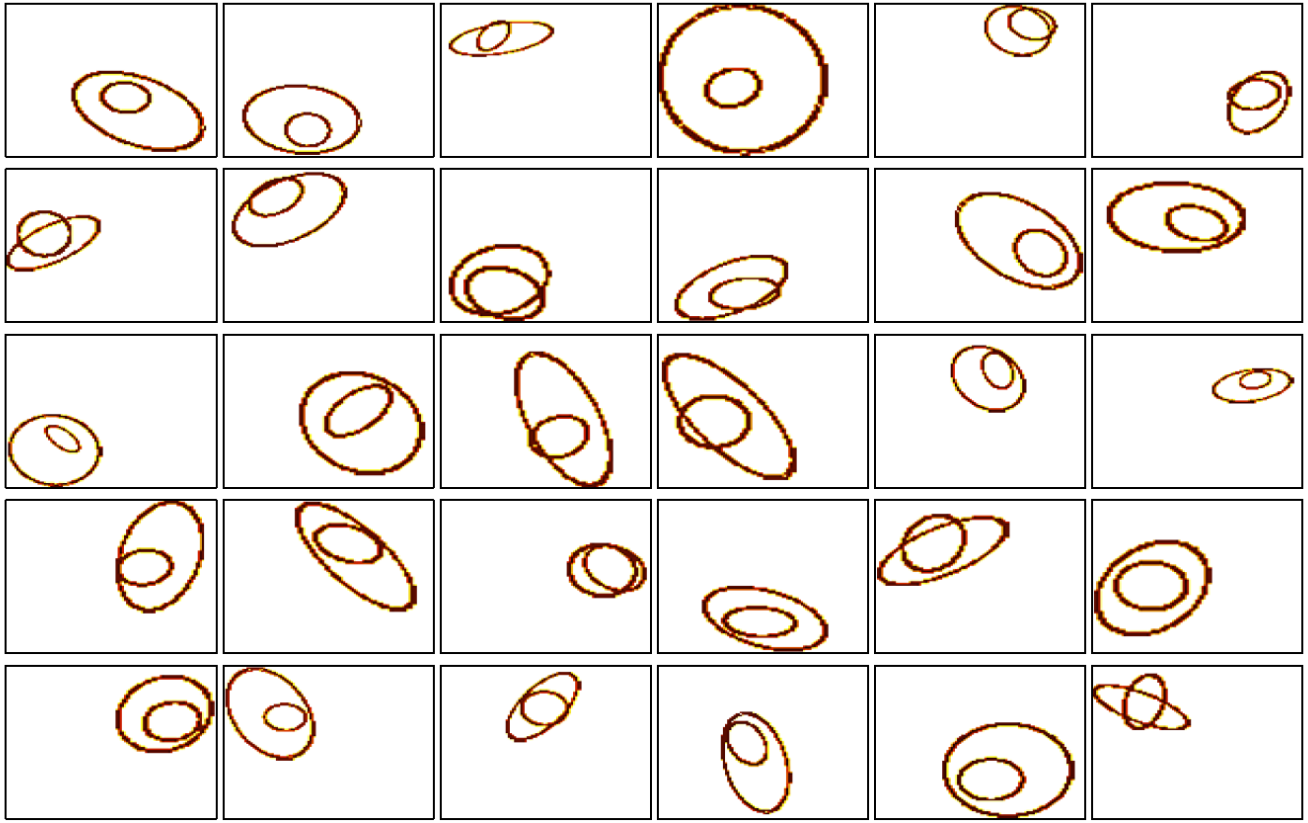


Figure: 30 artificial images of two nested random ellipses.

Wasserstein Barycenter Application

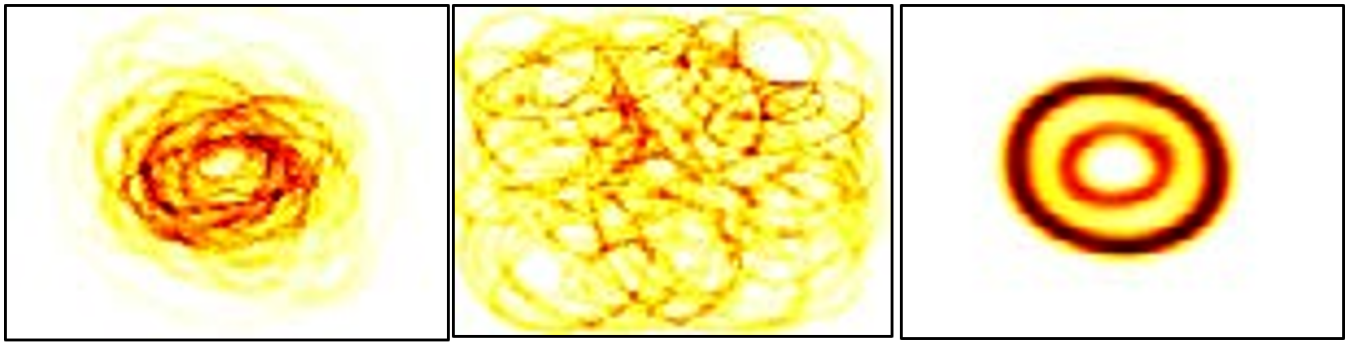


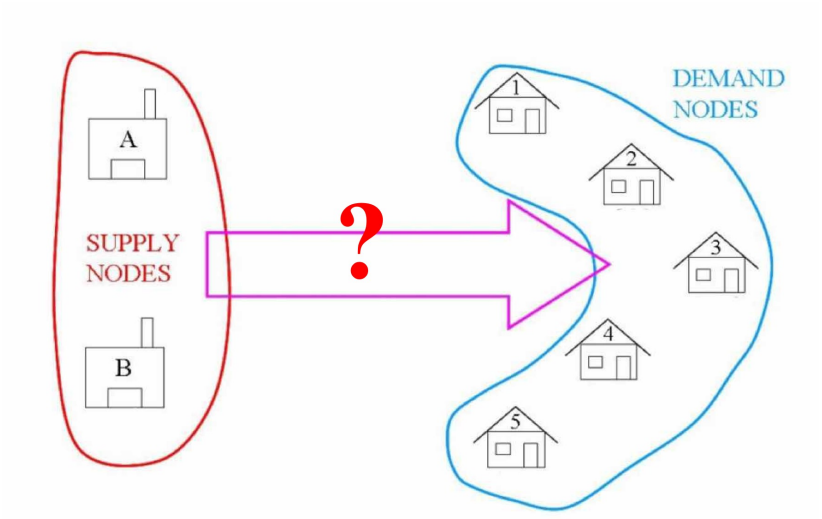
Figure: Mean measures using the (a) Euclidean after re-centering images (b) Euclidean distance (c) 2-Wasserstein distance

Wasserstein Barycenter Application:

- **Clustering**
  [3, N. Ho et al., 2017]
- **Image processing**
  [7, J. Rabin et al., 2012]
- **Others**
  [5, J. Lee et al., 2018]
  [4, M. J. Kusner et al., 2015]

Wasserstein Distance and Optimal Transportation Problem

Figure: Transportation problem



- m_s supply nodes are located at $\boldsymbol{q}_i^{(s)}$, and supply capacity is $\boldsymbol{a}^{(s)}$.
- m_d demand nodes are located at $\boldsymbol{q}_j^{(d)}$, and demands are $\boldsymbol{a}^{(d)}$.

Wasserstein Distance

2-Wasserstein distance $\mathcal{W}_2(\mathcal{P}^{(supply)}, \mathcal{P}^{(demand)})$

$$min\left\{\sqrt{\sum_{i=1}^{m_s}\sum_{j=1}^{m_d}\pi_{ij}\|\boldsymbol{q}_i^{(s)}-\boldsymbol{q}_j^{(d)}\|^2} : \Pi=[\pi_{ij}]\in\mathcal{M}(\boldsymbol{a}^{(s)},\boldsymbol{a}^{(d)})\right\}$$

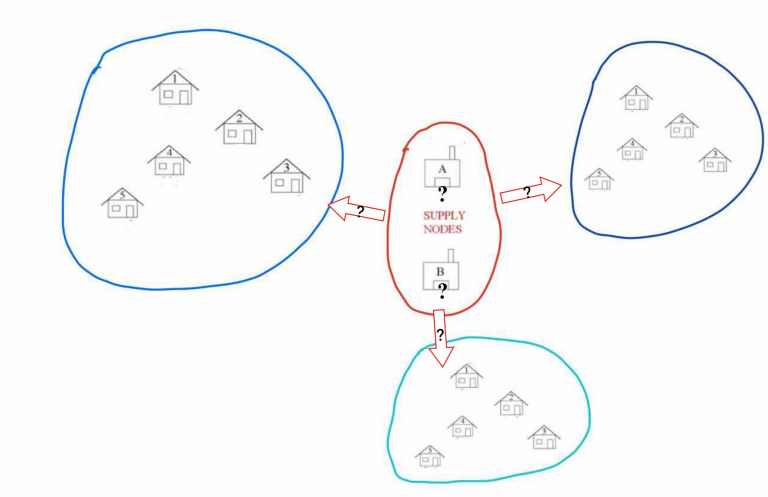
- Probability distributions: (locations and capacities)

$$\mathcal{P}^{(t)}=\{(a_i^{(t)},\boldsymbol{q}_i^{(t)}) : i=1,\cdots,m_t\},\quad \sum_{i=1}^{n}a_i^{(t)}=1, t=s,d.$$

- Measures with marginals $\boldsymbol{a}^{(s)}, \boldsymbol{a}^{(d)}$: (transportation plan)
$\mathcal{M}(\boldsymbol{a}^{(s)},\boldsymbol{a}^{(d)}) := \{\Pi\in\mathbb{R}_+^{m_s\times m_d} : \Pi\mathbf{1}_{m_s}=\boldsymbol{a}^{(s)}, \Pi^\top\mathbf{1}_{m_d}=\boldsymbol{a}^{(d)}\}$

Wasserstein Barycenter

Figure: If we don't know which group to serve



- N different markets $\{\mathcal{P}^{(t)}\}_{t=1}^{N}$
- **How to decide the locations and manage the inventory?**

Wasserstein Barycenter

Definition (Wasserstein Barycenter)

Wasserstein barycenter of N probability distributions $\left\{\mathcal{P}^{(t)}\right\}_{t=1}^{N}$ is the minimizer $\mathcal{P}^*$,

$$\arg\min_{\mathcal{P}} \frac{1}{N} \sum_{i=1}^{N} W_2^2 \left(\mathcal{P}, \mathcal{P}^{(i)}\right) \tag{1}$$

Wasserstein Barycenter

Main Problem

$$\begin{aligned} \min_{\boldsymbol{w}, X, \{\Pi^{(t)}\}} \quad & \sum_{t=1}^{N} \left\langle \mathcal{D}(X, Q^{(t)}), \Pi^{(t)} \right\rangle \\ \text{s.t.} \quad & \Pi^{(t)} \mathbf{1}_{m_t} = \boldsymbol{w}, \left(\Pi^{(t)}\right)^{\top} \mathbf{1}_m = \boldsymbol{a}^{(t)}, \Pi^{(t)} \geq 0, \forall t \\ & \mathbf{1}_m^{\top} \boldsymbol{w} = 1, \boldsymbol{w} \geq 0 \end{aligned} \tag{2}$$

- $\langle X, Q \rangle = \sum_{ij} X_{ij} Q_{ij}$
- Weight: $\boldsymbol{w} := (w_1, \cdots, w_m)^{\top} \in R_+^m$,
- Support points: $X := [\boldsymbol{x}_1, \cdots, \boldsymbol{x}_m] \in \mathbb{R}^{d \times m}$
- Measures with marginals: $\Pi^{(t)} \in R_+^{m \times m_t}$ for $t = 1, \cdots, N$
- Euclidean distances: $\mathcal{D}(X, Q^{(t)}) := [\|\boldsymbol{x}_i - \boldsymbol{q}_j^{(t)}\|^2] \in \mathbb{R}^{m \times m_t}$ for $t = 1, \cdots, N$

Mainstream Method: Sinkhorn Based Methods

- Iterative Bregman Projections by [1, J.D. Benamou et al., 2015]
- Smooth the problem with an entropic regularization term
- **Advantages:** fast, low memory usage
- **Disadvantages:** low accuracy

Figure: Representative Sinkhorn based method IBP(left) and our MAAIPM(right)

Traditional Algorithm

Result: X and $\boldsymbol{w}$

while *a termination criterion is not hold* **do**

Step 1. $\{\boldsymbol{w}^{(i+1)}, \Pi^{(i+1)}\} = \arg\min_{w,\Pi} \sum_{t=1}^{N} \left\langle \mathcal{D}(X^{(i+1)}, Q^{(t)}), \Pi^{(t)} \right\rangle$;

Step 2. $X^{(i+1)} = \arg\min_{X} \sum_{t=1}^{N} \left\langle \mathcal{D}(X, Q^{(t)}), \Pi^{(i,t)} \right\rangle$;

end

Algorithm 1: The Free Support Point Case

Step 1. Update $\{\boldsymbol{w}^{(i+1)}, \Pi^{(i+1)}\}$

The problem in step 1 can be reduced to a linear program

$$\begin{aligned} \min_{\boldsymbol{w},\{\Pi^{(t)}\}} \quad & \sum_{t=1}^{N} \left\langle D^{(t)}, \Pi^{(t)} \right\rangle \\ \text{s.t.} \quad & \Pi^{(t)} \mathbf{1}_{m_t} = \boldsymbol{w}, \left(\Pi^{(t)}\right)^{\top} \mathbf{1}_m = \boldsymbol{a}^{(t)}, \Pi^{(t)} \geq 0, \forall t = 1, \cdots, N \\ & \mathbf{1}_m^{\top} \boldsymbol{w} = 1, \boldsymbol{w} \geq 0 \end{aligned} \tag{3}$$

where $D^{(t)}$ denotes $\mathcal{D}(X, Q^{(t)})$ for simplicity.

Step 2. Update $X^{(i+1)}$

Proposition

With w and Π fixed, the objective value of original problem (2) is quadratic in terms of X, and the optimal X is solved by:

$$\boldsymbol{x}_i := \frac{1}{Nw_i} \sum_{t=1}^{N} \sum_{j=1}^{m_t} \pi_{ij}^{(t)} \boldsymbol{q}_j^{(t)} \tag{4}$$

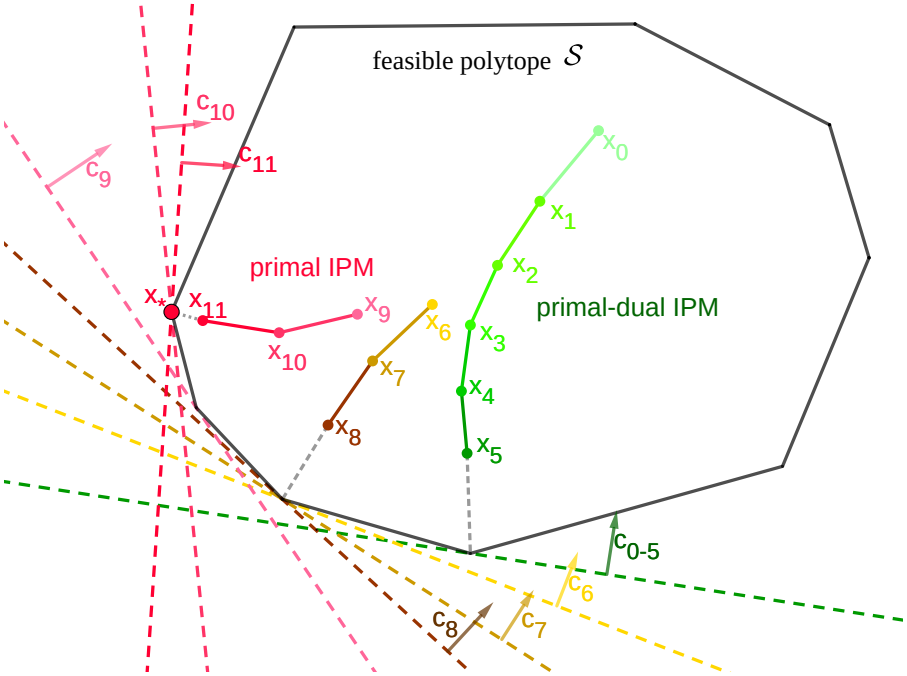
Proposition

In another word, the original problem(2) can be reformulated as

$$\min \boldsymbol{c}(\boldsymbol{x})^\top \boldsymbol{x} \quad s.t. A\boldsymbol{x} = \boldsymbol{b}, \ \boldsymbol{x} \geq 0. \tag{5}$$

Matrix-based Adaptive Alternating IPM(**MAAIPM**)

Figure: **MAAIPM**



- x_i is returned by each IPM iteration under the **objective gradient** c_i; and c_i is calculated by x_{i-1};
- At the beginning, c_i is updated after every a few **primal-dual IPM**[6] iterations(green).
- Then **primal IPM**(yellow and red) is applied to frequently update c and uses "jump" tricks to escape local minima.

Linear Programming

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ s.t. \quad & A\boldsymbol{x} = \boldsymbol{b} \quad \boldsymbol{x} \geq 0 \end{aligned}$$

$\boldsymbol{x} = (vec(\Pi^{(1)}); ...; vec(\Pi^{(N)}); \boldsymbol{w})$ and
$b = (\boldsymbol{a}^{(1)}; ...; \boldsymbol{a}^{(N)}; \mathbf{0}_m; ...; \mathbf{0}_m; 1)$

$$\text{and } A = \begin{bmatrix} I_{m_1} \otimes \mathbf{1}_m^T & & & \\ & \ddots & & \\ & & I_{m_N} \otimes \mathbf{1}_m^T & \\ \mathbf{1}_{m_1}^T \otimes I_m & & & -I_m \\ & \ddots & & \vdots \\ & & \mathbf{1}_{m_N}^T \otimes I_m & -I_m \\ & & & \mathbf{1}_m^T \end{bmatrix} \tag{6}$$

$n_{row} := Nm + \sum_{i=1}^N m_i + 1$ and $n_{col} := m \sum_{i=1}^N m_i + m$

Interior-Point Methods

- Primal problem:
  $\min c^T x, \quad \text{subject to } Ax = b, x \geq 0$
- Dual problem:
  $\max b^T \lambda, \quad \text{subject to } A^T \lambda + s = c, s \geq 0$
- **Main difficulty:** a huge System of linear equations

$$\left(AD^2A^\top\right)\Delta\lambda = \tilde{b}$$

, where

$$D^2 = \begin{pmatrix} s_1^{-1}x_1 & & & \\ & s_2^{-1}x_2 & & \\ & & \ddots & \\ & & & s_n^{-1}x_n \end{pmatrix}$$

A special case: $N = 3, m = 3, m_1 = 2, m_2 = m_3 = 3$

$$
A = \left[\begin{array}{cccccc|ccccccccc|ccccccccc|ccc}
1 & 1 & 1 & 0 & 0 & 0 & & & & & & & & & & & & & & & & & & & \\
0 & 0 & 0 & 1 & 1 & 1 & & & & & & & & & & & & & & & & & & & \\
\hline
 & & & & & & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & & & & & & & & & & \\
 & & & & & & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & & & & & & & & & & \\
 & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & & & & & & & & & & \\
\hline
 & & & & & & & & & & & & & & & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \\
 & & & & & & & & & & & & & & & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & \\
 & & & & & & & & & & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & \\
\hline
1 & 0 & 0 & 1 & 0 & 0 & & & & & & & & & & & & & & & & & & & -1 & 0 & 0 \\
0 & 1 & 0 & 0 & 1 & 0 & & & & & & & & & & & & & & & & & & & 0 & -1 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & & & & & & & & & & & & & & & & & & & 0 & 0 & -1 \\
\hline
 & & & & & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & & & & & & & & & & -1 & 0 & 0 \\
 & & & & & & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & & & & & & & & & & 0 & -1 & 0 \\
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\hline
 & & & & & & & & & & & & & & & & & & & & & & & & 1 & 1 & 1
\end{array}\right]
$$

A special case: $N = 3, m = 3, m_1 = 2, m_2 = m_3 = 3$

$$\exists L,\ s.t.\ L \cdot A =$$

$$\left[\begin{array}{cccccc|ccccccccc|ccccccccc|ccc}
1 & 1 & 1 & 0 & 0 & 0 & & & & & & & & & & & & & & & & & & & \\
0 & 0 & 0 & 1 & 1 & 1 & & & & & & & & & & & & & & & & & & & \\
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 & & & & & & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & & & & & & & & & & \\
 & & & & & & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & & & & & & & & & & \\
 & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & & & & & & & & & & \\
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 & & & & & & & & & & & & & & & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & \\
 & & & & & & & & & & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & \\
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0 & 0 & 0 & 0 & 0 & 0 & & & & & & & & & & & & & & & & & & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 1 & 0 & & & & & & & & & & & & & & & & & & 0 & -1 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & & & & & & & & & & & & & & & & & & 0 & 0 & -1 \\
\hline
 & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & & & & & & & & & 0 & 0 & 0 \\
 & & & & & & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & & & & & & & & & 0 & -1 & 0 \\
 & & & & & & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & & & & & & & & 0 & 0 & -1 \\
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\hline
 & & & & & & & & & & & & & & & & & & & & & & & & 1 & 1 & 1
\end{array}\right]$$

A special case: $N = 3, m = 3, m_1 = 2, m_2 = m_3 = 3$

Let $\bar{A} = L \cdot A$, and $\bar{A} D^2 \bar{A}^T =$

$$
\left[\begin{array}{cc|ccc|ccc|cc|cc|c}
\times & & & & & & \times & \times & & & & & \\
 & \times & & & & & \times & \times & & & & & \\
\hline
 & & \times & & & & & & & \times & \times & & \\
 & & & \times & & & & & & \times & \times & & \\
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\hline
 & & & & & \times & & & & & & \times & \times \\
 & & & & & & \times & & & & & \times & \times \\
 & & & & & & & \times & & & & \times & \times \\
\hline
\times & \times & & & & & \times & & y_1 & & y_1 & & -y_1 \\
\times & \times & & & & & & \times & & y_2 & & y_2 & -y_2 \\
\hline
 & & \times & \times & \times & & y_1 & & \times & & y_1 & & -y_1 \\
 & & \times & \times & \times & & & y_2 & & \times & & y_2 & -y_2 \\
\hline
 & & & & & \times & \times & \times & y_1 & & y_1 & & -y_1 \\
 & & & & & \times & \times & \times & & y_2 & & y_2 & -y_2 \\
\hline
 & & & & & & -y_1 & -y_2 & -y_1 & -y_2 & -y_1 & -y_2 & c
\end{array}\right]
$$

A special case: $N = 3, m = 3, m_1 = 2, m_2 = m_3 = 3$

$$\exists V, \quad s.t. V\bar{A}D^2\bar{A}^T V^T =$$

$$\left[\begin{array}{cc|ccc|cc|cc|cc|c}
\times & & & & & & & & & & & \\
 & \times & & & & & & & & & & \\
\hline
 & & \times & & & & & & & & & \\
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\hline
 & & & & & \times & \times & a & & a & & \\
 & & & & & \times & \times & & b & & b & \\
\hline
 & & & & & a & & \times & \times & a & & \\
 & & & & & & b & \times & \times & & b & \\
\hline
 & & & & & a & & a & & \times & \times & \\
 & & & & & & b & & b & \times & \times & \\
\hline
 & & & & & & & & & & & \times
\end{array}\right]$$

Low-Rank Regularization Method

Set $V\bar{A}D^2\bar{A}^TV^T$

$$\begin{bmatrix} B_1 & & \\ & A_1 + A_2 & \\ & & c \end{bmatrix}$$

In the aforementioned example,

$$A_2 = \begin{bmatrix} a & & a & & a & \\ & b & & b & & b \\ a & & a & & a & \\ & b & & b & & b \\ a & & a & & a & \\ & b & & b & & b \end{bmatrix}$$

Low-Rank Regularization Method

Let $A_2 = (\mathbf{1}_N \mathbf{1}_N^\top) \otimes (R^\top R)$ and $A_1 = diag(A_{11}, A_{22}, \cdots, A_{nn})$.
Use low-rank regularization:

$$(A_1 + A_2)^{-1} = A_1^{-1} - A_1^{-1} U \bar{R} (I + \bar{R}^\top U^\top A_1^{-1} U \bar{R})^{-1} \bar{R}^\top U^\top A_1^{-1},$$

where

$$\bar{R} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ R^\top \end{bmatrix}, \quad U = \begin{bmatrix} I_{m-1} & & & I_{m-1} \\ & I_{m-1} & & \vdots \\ & & \ddots & I_{m-1} \\ & & & I_{m-1} \end{bmatrix}.$$

Moreover,

$$\bar{R}(I + \bar{R}^\top U^\top A_1^{-1} U \bar{R})^{-1} \bar{R}^\top = (\boldsymbol{e}_N \boldsymbol{e}_N^\top) \otimes \left(R^\top (I + R(\sum_{i=1}^{N} A_{ii}^{-1}) R^\top)^{-1} R \right)$$

where $\mathbf{e}_N = (0, 0, \cdots, 1)^\top$.

Low-Rank Regularization Method

Remark

- The main computational cost of this procedure lies in computing $A_{ii}^{-1}, i = 1, .., N$, which is suitable for **parallel computing**
- When $m = m_t$, the **time complexity** of our algorithm in each iteration is $O(Nm^3)$, instead of standard IPM's complexity $O(N^3m^4)$
- When the support points of distributions are different, the **memory usage** is within a constant multiple of that of the first order method IBP

Solving the Normal Equations

- In practice, we develop Single low-rank regularization method **(SLRM)** and Double low-rank regularization method **(DLRM)** to solve the normal equations.
- SLRM is linear with respect to N and m_t, and DLRM is linear with respect to N and m.

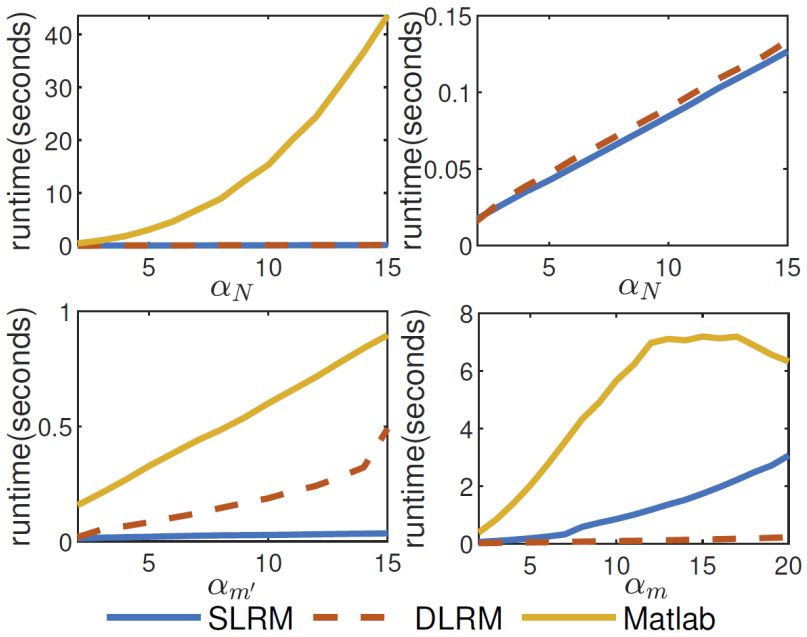


Figure: Rescaling N, m and m_t by respectively α_N, α_m and $\alpha_{m'}$ times.

Pre-specified Support Case

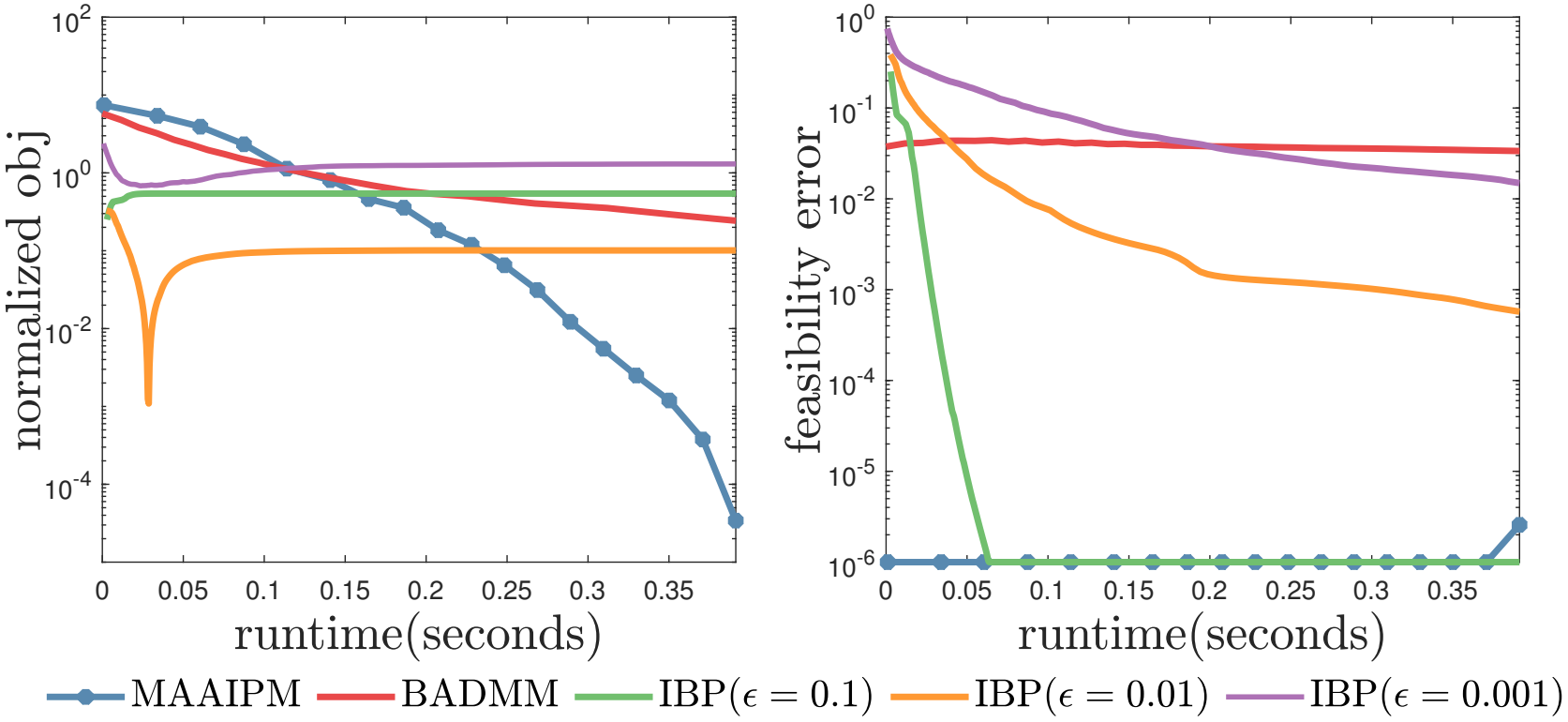


Figure: $N = m = m_t = 50$. "Normalized obj" measures the difference with the real optimal value. "Feasibility error" measures the distance to the feasible set.

Free Support Case

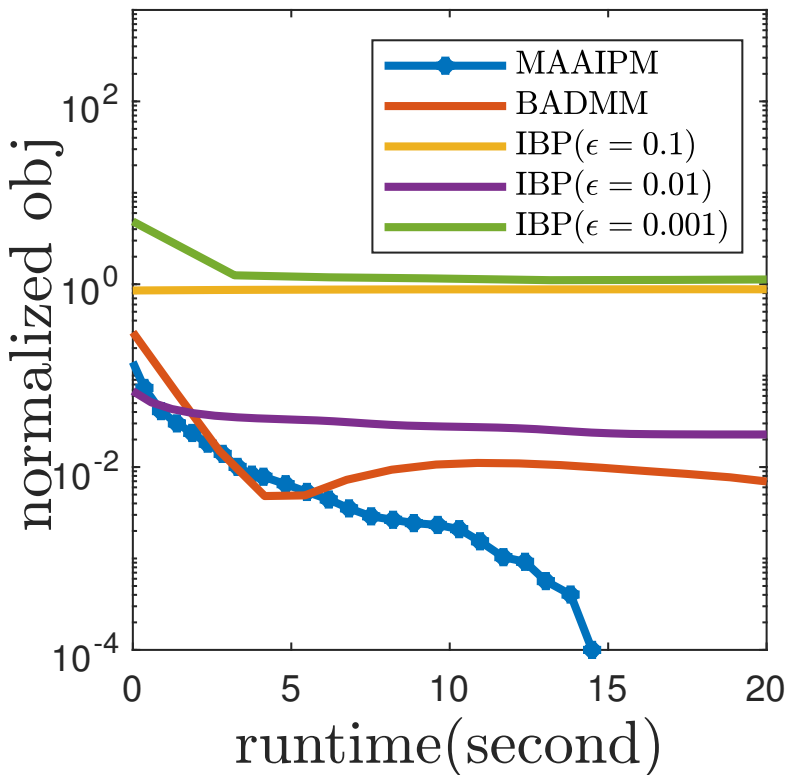
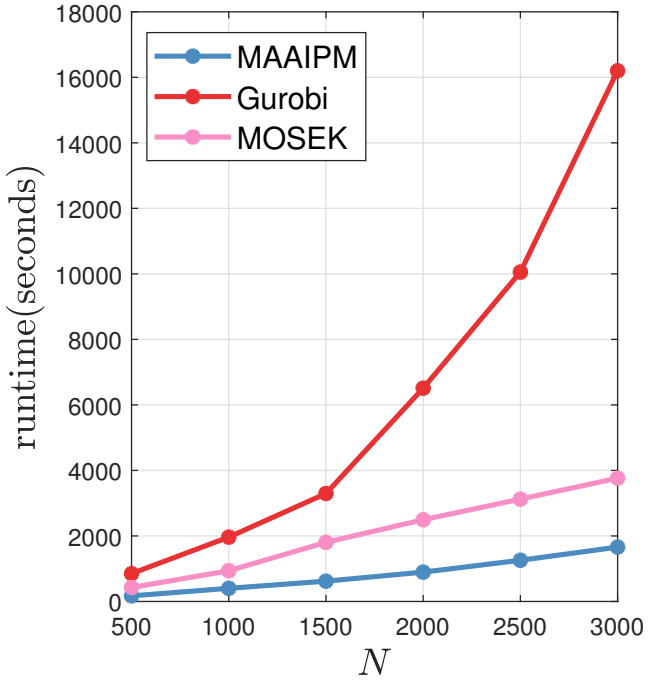


Figure: $N = m = m_t = 50$. "Normalized obj" denotes $\mathcal{F}_{method}/\mathcal{F}_{MAAIPM} - 1$, where $\mathcal{F}_{method}$ is the objective value obtained by each iteration of methods.

Low Memory Usage

Figure: $m_t \sim Unif(150, 250)$, $m = 200$.



- When $N = 5000$, MAAIPM requires only 3098.23 seconds, while MOSEK crashes after 23028.23 seconds.
- MAAIPM is approximately linear with respect to N and m(or m_t).

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